Game Playing

Adversarial search

Adversarial search

- **Adversarial search is a search, where we examine the problem which arises when we try to plan ahead of the world and other agents are planning against us.**

Game

- **Searches in which two or more players with conflicting goals are trying to explore the same search space for the solution, are called adversarial searches, often known as Games**.

- there might be some situations where more than one agent is searching for the solution in the same search space, and this situation usually occurs in game playing.

Types of Games in AI:

- **Perfect information:** A game with the perfect information is that in which agents can look into the complete board. Agents have all the information about the game, and they can see each other moves also. **Examples** are Chess, Checkers, etc.
- **Imperfect information:** If in a game agents do not have all information about the game and not aware with what's going on, such type of games are called the game with imperfect information, such as tic-tac-toe, Battleship, etc.

- **Deterministic games:** Deterministic games are those games which follow a strict pattern and set of rules for the games, and there is no randomness associated with them. Examples are chess, tic-tac-toe, etc.
- **Non-deterministic games:** Non-deterministic are those games which have various unpredictable events and has a factor of chance or luck. This factor of chance or luck is introduced by either dice or cards. These are random, and each action response is not fixed. Such games are also called as stochastic games.

  Example: Backgammon, Monopoly, Poker, etc.

Types of games

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	deterministic	**Chance**
Perfect information	Chess, rock-paper-scissors	Monopoly
Imperfect information	Battleships, tic-tac-toe	Poker ,scrabble

Zero-Sum Game

- Zero-sum games are adversarial search which involves pure competition.
- In Zero-sum game each agent's gain or loss of utility is exactly balanced by the losses or gains of utility of another agent.
- One player of the game try to maximize one single value, while other player tries to minimize it.
- Each move by one player in the game is called as **ply**.
- Chess and tic-tac-toe are examples of a Zero-sum game.

Zero-sum game: Embedded thinking

- The Zero-sum game involved embedded thinking in which one agent or player is trying to figure out:
- What to do.
- How to decide the move
- Needs to think about his opponent as well
- The opponent also thinks what to do
- Each of the players is trying to find out the response of his opponent to their actions. This requires embedded thinking or backward reasoning to solve the game problems in AI.

Formalization of the problem:

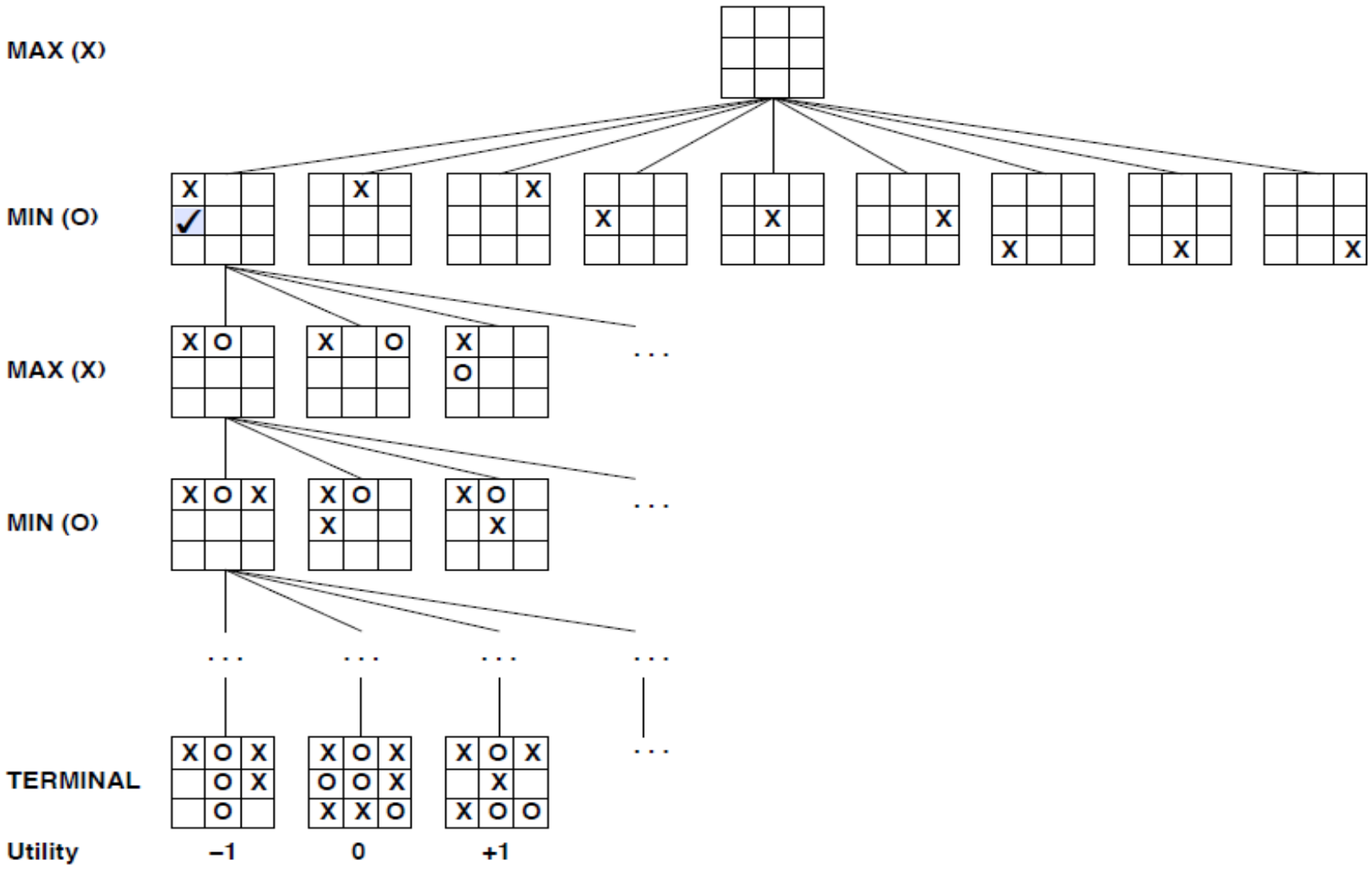
- **A game can be defined as a type of search in AI which can be formalized of the following elements:**
- **Initial state:** It specifies how the game is set up at the start.
- **Player(s):** It specifies which player has moved in the state space.
- **Action(s):** It returns the set of legal moves in state space.
- **Result(s, a):** It is the transition model, which specifies the result of moves in the state space.
- **Terminal-Test(s):** Terminal test is true if the game is over, else it is false at any case. The state where the game ends is called terminal states.
- **Utility(s, p):** A utility function gives the final numeric value for a game that ends in terminal states s for player p. It is also called payoff function. For Chess, the outcomes are a win, loss, or draw and its payoff values are +1, 0, ½. And for tic-tac-toe, utility values are +1, -1, and 0.

Game Tree

- A game tree is a tree where nodes of the tree are the game states and Edges of the tree are the moves by players. Game tree involves initial state, actions function, and result Function.

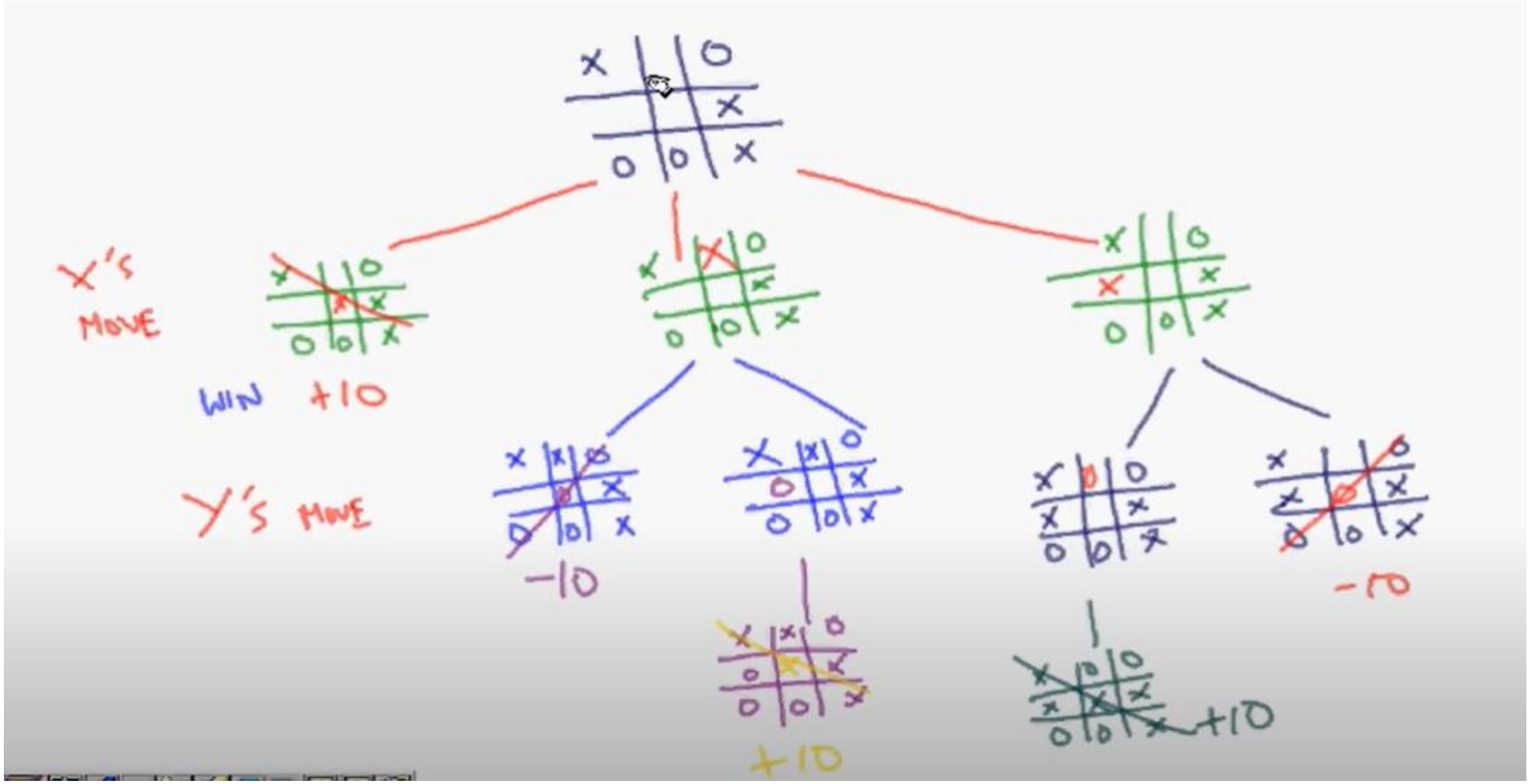
Example: Tic-Tac-Toe game tree:

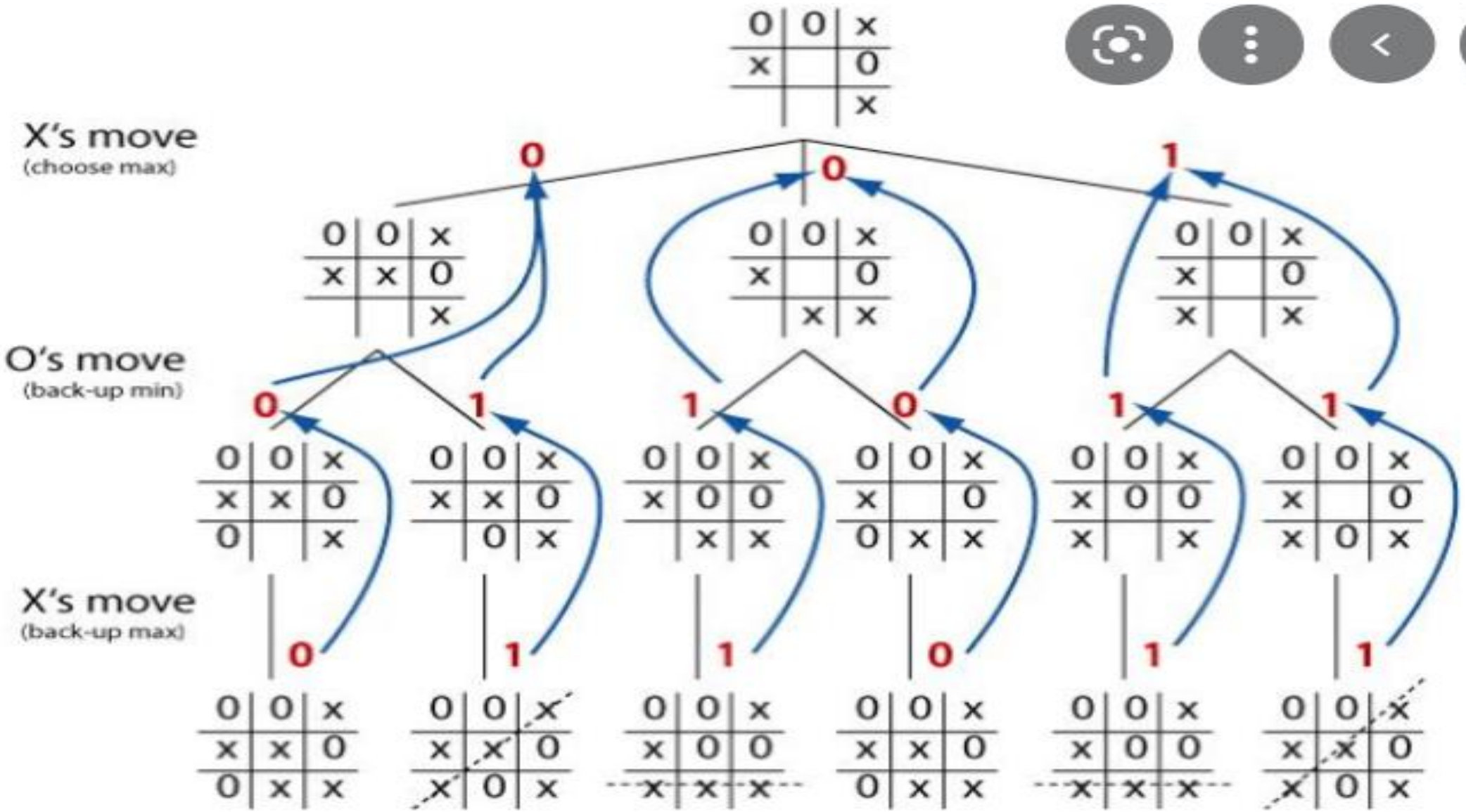
- The following figure is showing part of the game-tree for tic-tac-toe game. Following are some key points of the game:
- There are two players MAX and MIN.
- Players have an alternate turn and start with MAX.
- MAX maximizes the result of the game tree
- MIN minimizes the result.


MAX (X)
MIN (O)
MAX (X)
MIN (O)
TERMINAL
Utility
–1
0
+1

Minmax algorithm

- AIM: build an unbeatable tic-tac-toe
- An algorithm that calculates all the possible moves for the computer
- Tending to search for the optimal move for the winning of the player


X'S MOVE
WIN +10
Y'S MOVE
-10
+10
-10
+10


X's move
(choose max)
O's move
(back-up min)
X's move
(back-up max)

Alpha-Beta Pruning

- As we have seen in the minimax search algorithm that the number of game states it has to examine are exponential in depth of the tree.
- Since we cannot eliminate the exponent, but we can cut it to half.
- Hence there is a technique by which without checking each node of the game tree we can compute the correct minimax decision, and this technique is called **pruning**.
- This involves two threshold parameter Alpha and beta for future expansion, so it is called **alpha-beta pruning**.
- It is also called as **Alpha-Beta Algorithm**.

- The two-parameter can be defined as:
  - **Alpha:** The best (highest-value) choice we have found so far at any point along the path of Maximizer. The initial value of alpha is **-∞**.
  - **Beta:** The best (lowest-value) choice we have found so far at any point along the path of Minimizer. The initial value of beta is **+∞**.

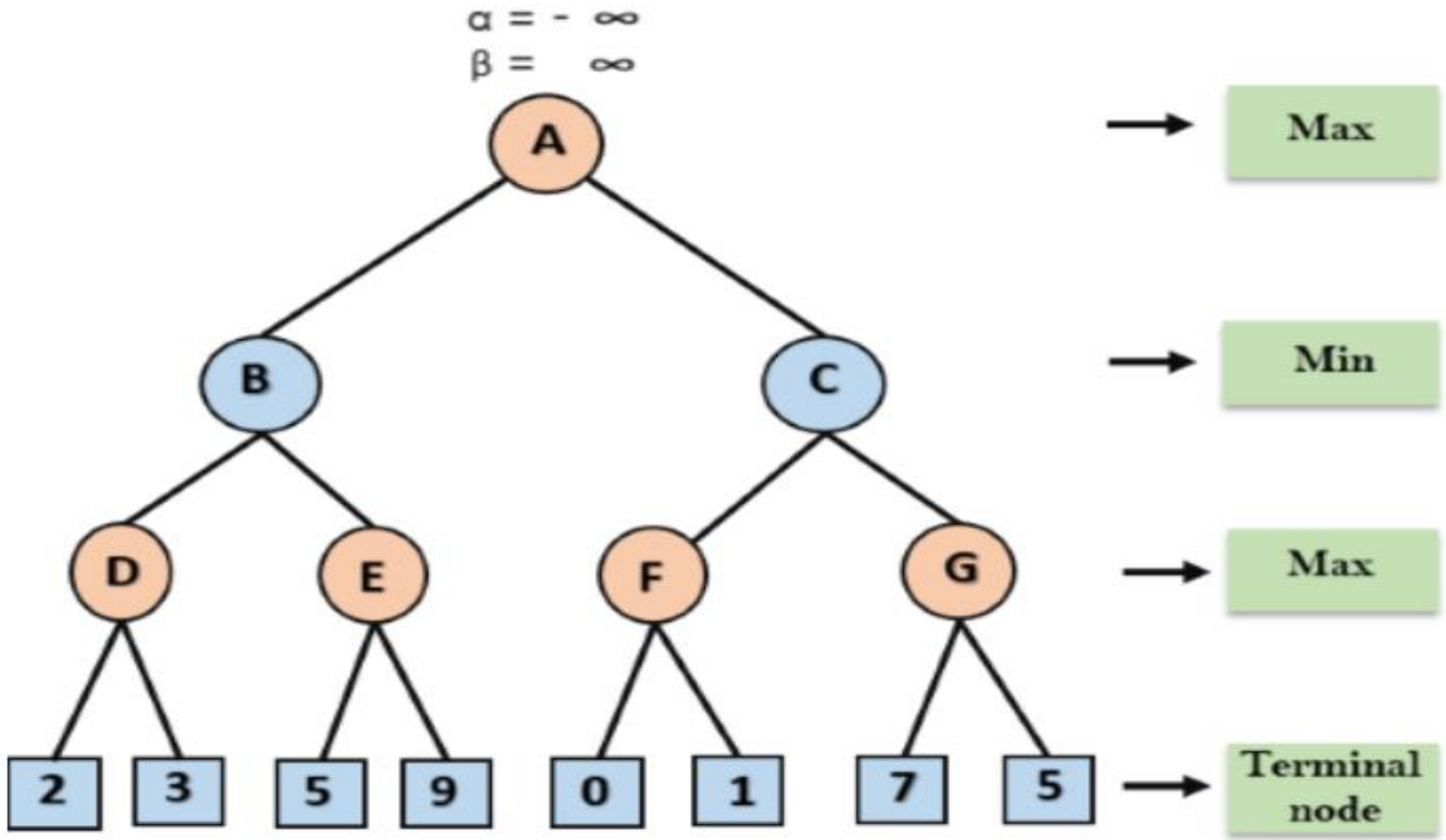
Condition

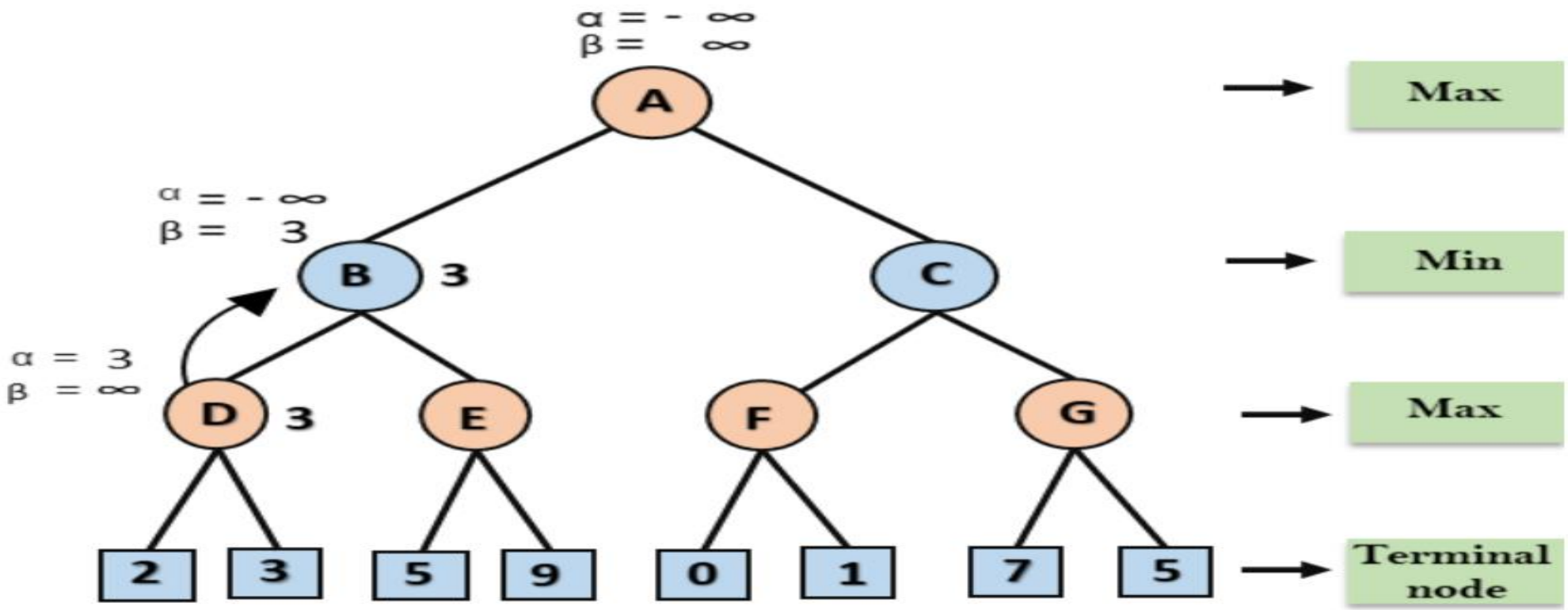
- The main condition which required for alpha-beta pruning is:

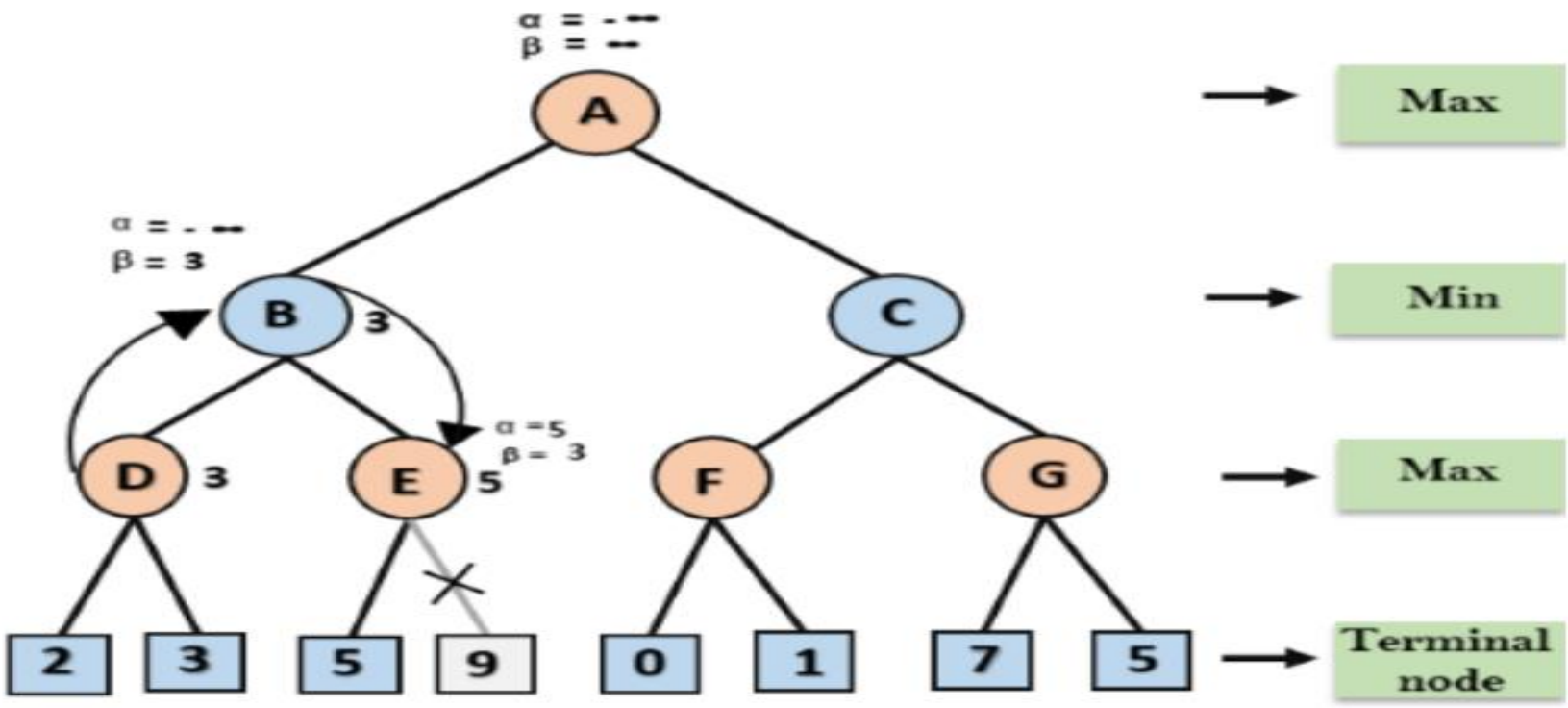
$$\alpha >= \beta$$

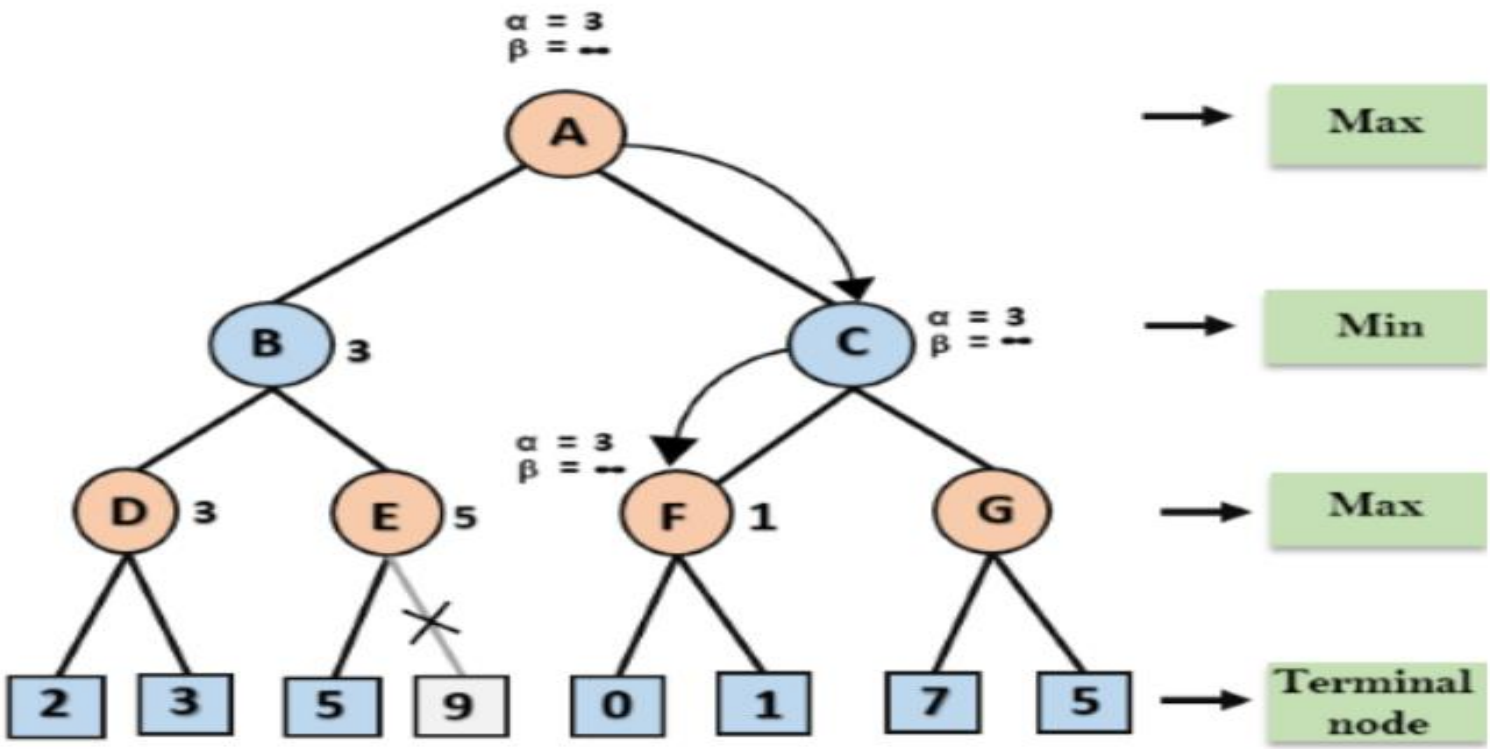
Key points about alpha-beta pruning:

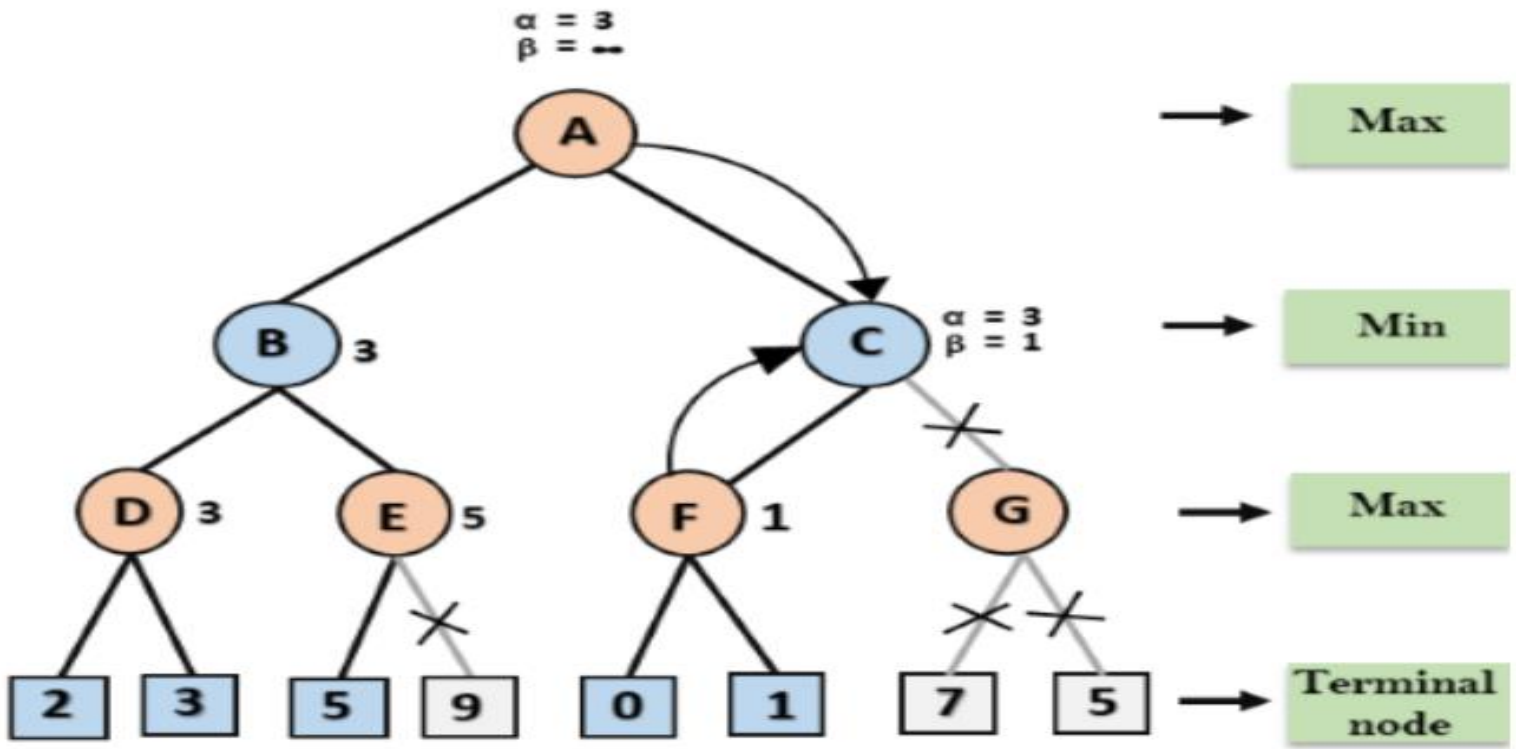
- The Max player will only update the value of alpha.
- The Min player will only update the value of beta.
- While backtracking the tree, the node values will be passed to upper nodes instead of values of alpha and beta.
- We will only pass the alpha, beta values to the child nodes.

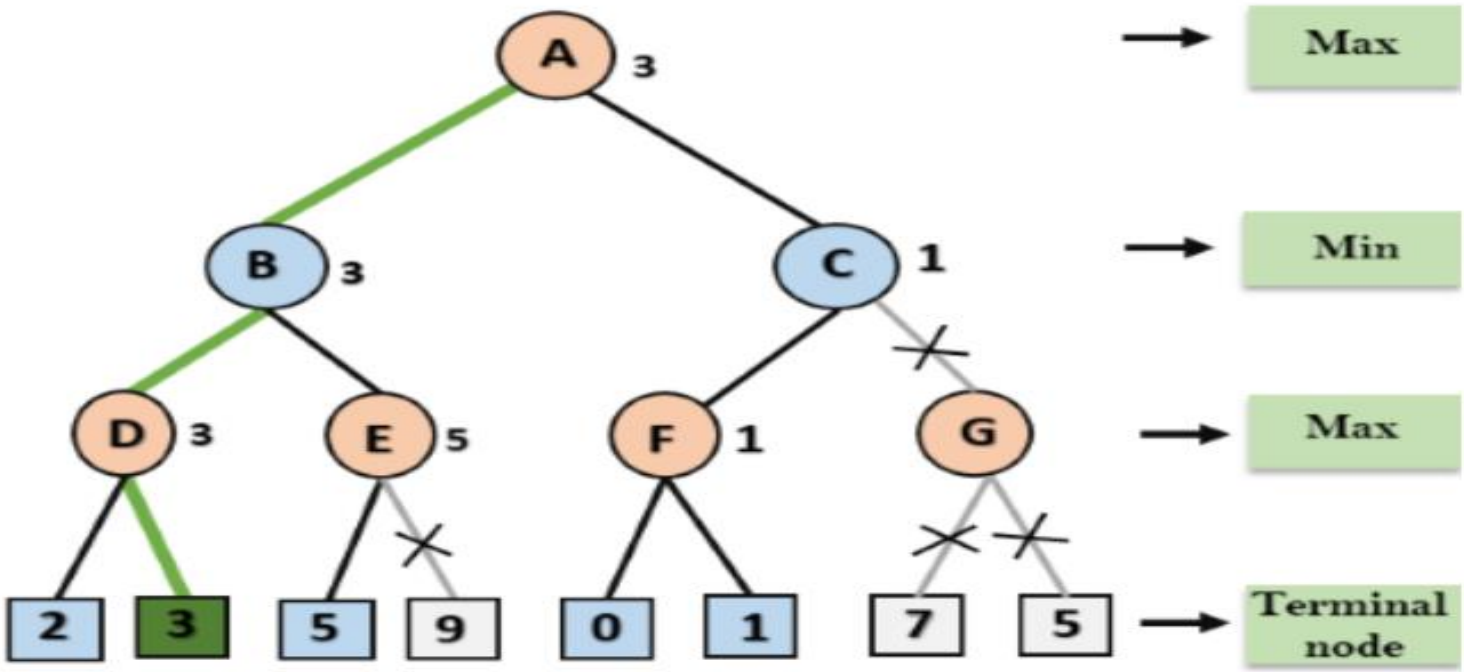

α = - ∞
β = ∞
A
B
C
D
E
F
G
2
3
5
9
0
1
7
5
Max
Min
Max
Terminal node


α = - ∞
β = ∞
A
Max
α = - ∞
β = 3
B
3
C
Min
α = 3
β = ∞
D
3
E
F
G
Max
2
3
5
9
0
1
7
5
Terminal node


α = -∞
β = ∞
A
Max
α = -∞
β = 3
B
3
C
Min
α =5
β = 3
D
3
E
5
F
G
Max
2
3
5
9
0
1
7
5
Terminal node


α = 3
β = --
A
B
3
C
α = 3
β = --
α = 3
β = --
D
3
E
5
F
1
G
2
3
5
9
0
1
7
5
Max
Min
Max
Terminal node


α = 3
β = ∞
A
B
3
C
α = 3
β = 1
D
3
E
5
F
1
G
2
3
5
9
0
1
7
5
Max
Min
Max
Terminal node


A
3
B
3
C
1
D
3
E
5
F
1
G
2
3
5
9
0
1
7
5
Max
Min
Max
Terminal node